

5.1.1 Task-Level Evidence: Testing the Framework’s Mechanisms in Our Data

We begin with direct evidence on the displacement and reallocation channels using task-level information extracted from our own job-postings data (Section 4.1). For each O*NET task j , firm i , 6-digit occupation c , and seniority group $s \in \{\text{junior}, \text{senior}\}$, let $\text{Prev}_{jics\tau}$ be the share of (i, c, s) postings in period τ whose extracted bundle contains task j . Splitting the sample at the ChatGPT launch ($\tau=\text{pre}$: 2021Q3–2022Q4; $\tau=\text{post}$: 2023Q1 through early April 2025, the latest postings in our extract), the dependent variable is the Davis et al. (1998) (DHS) growth rate:

$$\Delta^{\text{DHS}}\text{Prev}_{jics} = 2 \frac{\text{Prev}_{jics}^{\text{post}} - \text{Prev}_{jics}^{\text{pre}}}{\text{Prev}_{jics}^{\text{post}} + \text{Prev}_{jics}^{\text{pre}}} \times 100. \quad (7)$$

This is similar to the task-level DHS measure used by Hampshire et al. (2025). Observations are weighted by the harmonic mean of pre- and post-period cell sizes (the standard DHS weight), and standard errors are clustered at the firm level throughout.

Test 1: Displacement. Under the maintained assumption $\alpha < \hat{x}$, automated tasks lie in the junior range, so displacement predicts that AI-exposed tasks are removed from junior bundles at adopters post-ChatGPT and not from senior ones. We operationalize a task’s *displaceability* with the standardized Eloundou et al. (2024) GPT-4 exposure score Exp_j , and estimate, separately for juniors and seniors,

$$\Delta^{\text{DHS}}\text{Prev}_{jics} = \beta^s \text{Exp}_j \times \text{Adopt}_i + \alpha_j + \alpha_{ic} + \varepsilon_{jics}, \quad (8)$$

and a pooled triple-difference that adds the Senior_s dimension:

$$\Delta^{\text{DHS}}\text{Prev}_{jics} = \beta^A \text{Exp}_j \times \text{Adopt}_i + \beta^{AS} \text{Exp}_j \times \text{Adopt}_i \times \text{Senior}_s + \mathbf{\Gamma}'\mathbf{Z}_{jics} + \alpha_j + \alpha_{ic} + \varepsilon_{jics}, \quad (9)$$

where Adopt_i is the firm-level adopter indicator (Section 4.2), $\text{Senior}_s = 1$ for seniors, $\mathbf{Z}_{jics}$ collects the lower-order interactions, and α_j and α_{ic} are task and firm-by-occupation fixed effects (firm only in the less demanding spec). Identification comes from within-cell, cross-task variation in exposure interacted with adoption. Under displacement, we expect $\beta^J < 0$, $\beta^S \geq 0$, and $\beta^{AS} > 0$. Table 1 reports the results.

Table 1: Task-Level DHS Demand by GenAI Exposure and Seniority (Automation)

	Juniors only		Seniors only		Pooled	
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Exp}_j \times \text{Adopt}_i$	-3.923*	-4.140*	+2.408	+1.890	-4.152**	-3.667
	(2.065)	(2.347)	(3.203)	(3.128)	(1.842)	(2.394)
$\text{Exp}_j \times \text{Adopt}_i \times \text{Senior}_s$					+6.868**	+5.826*
					(3.490)	(3.237)
Task FE	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Occ FE		✓		✓		✓
Occ $\times$ Firm FE		✓		✓		✓
N	208,984	208,984	402,363	402,363	611,347	611,347
Firms	923	923	969	969	997	997

Notes: Unit of observation is (task $\times$ firm $\times$ 6-digit SOC occupation $\times$ seniority group). The dependent variable is $100 \times$ the Davis et al. (1998) (DHS) growth rate in the task's share of the cell's posting bundles, comparing post-ChatGPT (2023Q1 through early April 2025) to pre-ChatGPT (2021Q3–2022Q4). Exp_j is the standardized Eloundou et al. (2024) GPT-4 exposure of task j ; Adopt_i is the firm-level GenAI-adopter indicator (Section 4.2); Senior_s equals one for seniors. Observations are weighted by the harmonic mean of pre- and post-period cell sizes (the DHS weight). Standard errors in parentheses are clustered at the firm level. *, **, *** denote significance at 10%, 5%, and 1%.

For juniors, a one SD increase in a task's exposure is associated with a roughly four percentage-point larger DHS contraction in junior bundles at adopters relative to non-adopters, marginally significant in both specifications (columns 1–2). Scaling to the gap between the least- and most-exposed tasks (about four SDs) yields a 15–17 percentage-point differential. The coefficient for seniors (columns 3–4) is positive and insignificant: within the same firm-occupation cells, exposed tasks leave junior bundles but not senior ones. The pooled triple-difference makes the contrast precise: $\beta^A = -4.15$ ($p = 0.024$) and $\beta^{AS} = +6.87$ ($p = 0.049$) under firm fixed effects; the saturated specification delivers $\beta^{AS} = +5.83$ ($p = 0.072$). The contraction is therefore a within-occupation removal of GenAI-exposed tasks from junior bundles, not a compositional shift across occupations.

Test 2: Reallocation. In the framework, the reallocation channel works in the opposite direction: because GenAI augments juniors more than seniors ($\delta_j \geq \delta_s$), the post-AI allocation threshold satisfies $\hat{x}' \geq \hat{x}$, so tasks previously performed by seniors near the comparative-advantage boundary can now be absorbed by juniors. A direct test of this channel asks whether tasks that were ex-ante *senior-intensive* become more prevalent in

junior bundles at adopting firms post-ChatGPT.

We operationalize the model’s notion of an “ex-ante senior task” with a task-level senior-intensity score. For each O*NET task j , define

$$\text{SenInt}_j = \frac{\# \{(\text{posting}, j) \text{ pairs in pre-period at senior level}\}}{\# \{(\text{posting}, j) \text{ pairs in pre-period}\}}, \quad (10)$$

restricted to tasks appearing in at least 20 pre-period posting-task pairs (6,985 tasks), and standardized to unit variance. SenInt_j measures where, on the pre-AI seniority gradient, task j predominantly sat: tasks close to the boundary $\hat{x}$ on the senior side score high.

We re-estimate equations (8) and (9) with SenInt_j in place of Exp_j , and denote the resulting coefficients with tildes. Under reallocation, the model predicts $\tilde{\beta}^J > 0$ (senior-intensive tasks shift *into* junior bundles at adopters), $\tilde{\beta}^S \leq 0$ (the mirror image: those same tasks leave senior bundles), and in the pooled specification $\tilde{\beta}^A > 0$ with $\tilde{\beta}^{AS} < 0$. Table 2 reports the results.

Table 2: Task-Level DHS Demand by Senior-Intensity and Seniority (Reallocation)

	Juniors only		Seniors only		Pooled	
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{SenInt}_j \times \text{Adopt}_i$	−1.114 (2.995)	−2.097 (3.384)	−0.495 (3.284)	−1.657 (3.034)	−0.012 (3.401)	−1.783 (3.407)
$\text{SenInt}_j \times \text{Adopt}_i \times \text{Senior}_s$					−0.835 (5.031)	−0.468 (4.354)
Task FE	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Occ FE		✓		✓		✓
Occ $\times$ Firm FE		✓		✓		✓
N	189,495	189,495	377,846	377,846	567,341	567,341
Firms	923	923	969	969	997	997

Notes: Unit of observation is (task $\times$ firm $\times$ 6-digit SOC occupation $\times$ seniority group). The dependent variable is $100 \times$ the Davis et al. (1998) (DHS) growth rate in the task’s share of the cell’s posting bundles, comparing post-ChatGPT (2023Q1 through early April 2025) to pre-ChatGPT (2021Q3–2022Q4). SenInt_j is the task-level senior-intensity score defined in equation (10), standardized to unit variance and restricted to tasks with at least 20 pre-period (posting, task) pairs (6,985 tasks). Columns (5)–(6) include the lower-order terms Senior_s , $\text{SenInt}_j \times \text{Senior}_s$, and $\text{Adopt}_i \times \text{Senior}_s$ (coefficients not reported). Observations are weighted by the harmonic mean of pre- and post-period cell sizes (the DHS weight). Standard errors in parentheses are clustered at the firm level. *, **, *** denote significance at 10%, 5%, and 1%.

The pattern is uniformly null. None of the six coefficients on $\text{SenInt}_j \times \text{Adopt}_i$ is sta-

tistically distinguishable from zero, and all point estimates are small and—if anything—slightly *negative*, the wrong sign for the reallocation prediction. The senior–junior differential $\tilde{\beta}^{AS}$ in columns (5)–(6) is similarly close to zero (-0.84 and -0.47 , $p > 0.86$). To rule out the possibility that the null is masked by mechanical correlation between SenInt_j and Exp_j , we residualize SenInt_j on Exp_j at the task level—isolating senior-intensity orthogonal to AI-displaceability—and re-estimate the most demanding firm-by-occupation specification. The coefficients are again indistinguishable from zero.

The two tests use only weakly correlated regressors ($\text{corr}(\text{Exp}_j, \text{SenInt}_j) = +0.16$) and examine opposite directions of task movement. The results support the displacement channel, which reduces demand for junior labor in the framework, but we find no support for the reallocation channel, which would raise demand for juniors. Together, these findings support the prediction that GenAI adoption reduces demand for junior labor, consistent with the firm-level employment results we document in Sections 5.2–5.5.

5.1.2 Evidence from the Anthropic Economic Index

We complement the tests above with external evidence on the framework’s maintained assumption that the bulk of automated tasks lie in the junior range ($\alpha < \hat{x}$).²⁵ We use the Anthropic Economic Index (Appel et al., 2026), which classifies real-world interactions with a frontier large language model by the O*NET task they address and the *collaboration mode* of the interaction. Following the AEI’s methodology, we treat directive interactions and feedback loops as *displacement-type* usage, and learning, task iteration, and validation as *augmentation-type* usage. We merge these AEI measures with occupation-level seniority constructed from the Revelio data, yielding a sample of 387 occupations with sufficient AEI coverage.

We find that *displacement intensity*—the share of classified interactions accounted for by displacement-type modes—exceeds 50% for junior-type occupations but falls to 40% for senior-type occupations, a gap driven by rising task iteration and validation with seniority. The pattern is robust to controlling for broad occupation groups: with two-

²⁵This concerns the *mode* of GenAI engagement across tasks, not the size of the within-task productivity gain: the model assumes $\delta_j \geq \delta_s$ (juniors gain at least as much productivity conditional on task survival), but implies that a larger *share* of senior-type tasks will involve augmentation-type interaction because fewer of those tasks are amenable to outright automation.